

where

$$D = L^\mu \nabla_\mu = \frac{\partial}{\partial r} \quad (3.28)$$

$$\Delta = n^\mu \nabla_\mu = V \frac{\partial}{\partial r} + \frac{\partial}{\partial u} + X^i \frac{\partial}{\partial x^i} \quad (3.29)$$

$$\mathcal{S} = m^\mu \nabla_\mu = W \frac{\partial}{\partial r} + \xi^i \frac{\partial}{\partial x^i} \quad (3.30)$$

$$\phi_{00} = -\frac{1}{2} R_{\quad 00} = \bar{\phi}_{00} \quad (3.31)$$

$$\phi_{11} = -\frac{1}{4} (R_{12} + R_{34}) = \bar{\phi}_{11} \quad (3.32)$$

$$\phi_{01} = -\frac{1}{2} R_{13} = \bar{\phi}_{10} \quad (3.33)$$

$$\phi_{12} = -\frac{1}{2} R_{23} = \bar{\phi}_{21} \quad (3.34)$$

$$\phi_{02} = -\frac{1}{2} R_{33} = \bar{\phi}_{20} \quad (3.35)$$

$$\phi_{22} = -\frac{1}{2} R_{22} = \bar{\phi}_{22} \quad (3.36)$$

$$\Lambda = \frac{R}{24} \quad (3.37)$$

$$\psi_0 = -C_{1313} = -C_{\alpha\beta\gamma\delta} L^\alpha m^\beta L^\gamma m^\delta \quad (3.38)$$

$$\psi_1 = -C_{1213} = -C_{\alpha\beta\gamma\delta} L^\alpha n^\beta L^\gamma m^\delta \quad (3.39)$$

$$\begin{aligned} \psi_2 = -\frac{1}{2} (C_{1212} + C_{1234}) &= -\frac{1}{2} C_{\alpha\beta\gamma\delta} \\ &\times (L^\alpha n^\beta L^\gamma n^\delta + L^\alpha n^\beta m^\gamma \bar{m}^\delta) \end{aligned} \quad (3.40)$$

$$\psi_3 = C_{1224} = C_{\alpha\beta\gamma\delta} L^\alpha n^\beta n^\gamma \bar{m}^\delta \quad (3.41)$$

$$\psi_4 = -C_{2424} = -C_{\alpha\beta\gamma\delta} n^\alpha \bar{m}^\beta n^\gamma \bar{m}^\delta \quad (3.42)$$

Expressing the rotation coefficients in terms of the metric,
we have:

$$D \xi^i = \rho \xi^i + \sigma \bar{\xi}^i \quad (3.43)$$

$$D \omega = \rho \omega + \sigma \bar{\omega} - (\bar{\alpha} + \beta) \quad (3.44)$$

$$D X^L = \tau \bar{\xi}^i + \bar{\tau} \xi^i \quad (3.45)$$

$$D U = \tau \bar{\omega} + \bar{\tau} \omega - (\gamma + \bar{\gamma}) \quad (3.46)$$

$$\delta X^i - \Delta \xi^i = (\mu + \bar{\gamma} - \gamma) \xi^i + \bar{\lambda} \bar{\xi}^i \quad (3.47)$$

$$\bar{\delta} \xi^i - \bar{\delta} \bar{\xi}^i = (\bar{\beta} - \alpha) \xi^i + (\bar{\alpha} - \beta) \bar{\xi}^i \quad (3.48)$$

$$\delta \bar{\omega} - \bar{\delta} \omega = (\bar{\beta} - \alpha) \omega + (\bar{\alpha} - \beta) \bar{\omega} + (\mu - \bar{\mu}) \quad (3.49)$$

$$\delta U - \Delta \omega = (\mu + \bar{\gamma} - \gamma) \omega + \bar{\lambda} \bar{\omega} - \bar{v} \quad (3.50)$$

As in Chapter 2 we use the Bianchi identities as field equations for the Weyl tensor. In the Newman-Penrose formalism they may be written:

(I am indebted to R. G. McLenaghan for these)

$$\bar{\delta} \psi_0 - \mathcal{D} \psi_1 + \mathcal{D} \phi_{01} - \delta \phi_{00} = 4\alpha \psi_0 - 4\rho \psi_1 - (2\bar{\alpha} + 2\beta) \phi_{00} + 2\rho \phi_{01} + 2\sigma \phi_{10} \quad (3.51)$$

$$\Delta \psi_0 - \delta \psi_1 + \mathcal{D} \phi_{02} - \delta \phi_{01} = (4\gamma - \mu) \psi_0 - 2(2\tau + \beta) \psi_1 + 3\sigma \psi_2 - \bar{\lambda} \phi_{00} - 2\beta \phi_{01} + 2\sigma \phi_{11} + \rho \phi_{02} \quad (3.52)$$

$$3(\bar{\delta} \psi_1 - \mathcal{D} \psi_2) + 2(\mathcal{D} \phi_{11} - \delta \phi_{10}) + \bar{\delta} \phi_{01} - \Delta \phi_{00} = 3\lambda \psi_0 - 9\rho \psi_2 + 6\alpha \psi_1 + (\bar{\mu} - 2\mu - 2\gamma - 2\bar{\gamma}) \phi_{00} + (2\alpha + 2\bar{\tau}) \phi_{01} + 2(\tau - 2\bar{\alpha}) \phi_{10} + 2\rho \phi_{11} + 2\sigma \phi_{20} - \bar{\sigma} \phi_{02} \quad (3.53)$$

$$3(\Delta \psi_1 - \delta \psi_2) + 2(\mathcal{D} \phi_{12} - \delta \phi_{11}) + (\bar{\delta} \phi_{02} - \Delta \phi_{01}) = 3r \psi_0 + 6(\gamma - \mu) \psi_1 - 9\tau \psi_2 + 6\sigma \psi_3 - \bar{\nu} \phi_{00} + 2(\bar{\mu} - \mu - \gamma) \phi_{01} - 2\lambda \phi_{10} + 2\bar{\tau} \phi_{11} + (2\alpha + \bar{\tau} - 2\bar{\beta}) \phi_{02} + 2\sigma \phi_{21} \quad (3.54)$$

$$3(\bar{\delta} \psi_2 - \mathcal{D} \psi_3) + (\mathcal{D} \phi_{21} - \delta \phi_{20}) + 2(\bar{\delta} \phi_{11} - \Delta \phi_{10}) = 6\lambda \psi_1 - 6\rho \psi_3 - 2r \phi_{00} + 2\lambda \phi_{01} + 2(\bar{\mu} - \mu - 2\bar{\gamma}) \phi_{10} + 4\bar{\tau} \phi_{11} + (2\beta + 2\bar{\tau} - 2\bar{\alpha}) \phi_{20} - 2\bar{\sigma} \phi_{12} \quad (3.55)$$

$$\begin{aligned}
& 3(\Delta\psi_2 - \delta\psi_3) + (D\phi_{22} - \delta\phi_{21}) + 2(\bar{\delta}\phi_{12} - \Delta\phi_{11}) \\
& = 6\nu\psi_1 - 9\mu\psi_2 + 6(\beta - \tau)\psi_3 + 3\sigma\psi_4 \\
& - 2\nu\phi_{01} - 2\bar{\nu}\phi_{10} + 2(2\bar{\mu} - \mu)\phi_{11} + 2\lambda\phi_{02} \\
& - \bar{\lambda}\phi_{20} + 2(\bar{\epsilon} - 2\bar{\beta})\phi_{12} + 2(\beta + \tau)\phi_{21} - \rho\phi_{22} \quad (3.56)
\end{aligned}$$

$$\begin{aligned}
& \bar{\delta}\psi_3 - D\psi_4 + \bar{\delta}\phi_{21} - \Delta\phi_{20} = 3\lambda\psi_2 - 2\alpha\psi_3 - \rho\psi_4 \\
& - 2\nu\phi_{01} + 2\lambda\phi_{11} + (2\gamma - 2\bar{\gamma} + \bar{\mu})\phi_{20} \\
& + 2(\bar{\epsilon} - \alpha)\phi_{21} - \bar{\sigma}\phi_{22} \quad (3.57)
\end{aligned}$$

$$\begin{aligned}
& \Delta\psi_3 - \delta\psi_4 + \bar{\delta}\phi_{22} - \Delta\phi_{21} = 3\nu\psi_2 - 2(\gamma + 2\mu)\psi_3 \\
& + (4\beta - \tau)\psi_4 - 2\nu\phi_{11} - \bar{\nu}\phi_{20} + 2\lambda\phi_{12} \\
& + 2(\gamma + \bar{\mu})\phi_{21} + (\bar{\epsilon} - 2\bar{\beta} - 2\alpha)\phi_{22} \quad (3.58)
\end{aligned}$$

$$\begin{aligned}
& D\phi_{12} - \delta\phi_{11} - \bar{\delta}\phi_{02} + \Delta\phi_{01} + 3\delta\Lambda = (2\gamma - \mu - 2\bar{\mu})\phi_{01} + \bar{\nu}\phi_{00} \\
& - \bar{\lambda}\phi_{10} - 2\tau\phi_{11} + (2\bar{\beta} - 2\alpha - \bar{\epsilon})\phi_{02} + 3\rho\phi_{12} + \sigma\phi_{21} \quad (3.59)
\end{aligned}$$

$$\begin{aligned}
& D\phi_{11} - \delta\phi_{10} - \bar{\delta}\phi_{01} + \Delta\phi_{00} + 3D\Lambda = (2\gamma - \mu + 2\bar{\gamma} - \bar{\mu})\phi_{00} - 2(\alpha + \bar{\epsilon})\phi_{01} \\
& - 2(\bar{\alpha} + \tau)\phi_{10} + 4\rho\phi_{11} + \bar{\sigma}\phi_{02} + \sigma\phi_{20} \quad (3.60)
\end{aligned}$$

$$\begin{aligned}
& D\phi_{22} - \delta\phi_{21} - \bar{\delta}\phi_{12} + \Delta\phi_{11} + 3\Delta\Lambda = \nu\phi_{01} + \bar{\nu}\phi_{10} - 2(\mu + \bar{\mu})\phi_{11} \\
& - \lambda\phi_{02} - \bar{\lambda}\phi_{20} + (2\bar{\beta} - \bar{\epsilon})\phi_{12} + (2\beta - \tau)\phi_{21} \\
& + 2\rho\phi_{22} \quad (3.61)
\end{aligned}$$

4. The Undisturbed Metric

The undisturbed metric may be written

$$ds^2 = \Omega^2 (dt^2 - d\rho^2 - \sinh^2 \rho (d\theta^2 + \sin^2 \theta d\phi^2))$$

$$\Omega = A(\cosh t - 1)$$

put $u = t - \rho$

then $ds^2 = \Omega^2 [-du^2 + 2dudt - \sinh^2(t-u)(d\theta^2 + \sin^2 \theta d\phi^2)]$ (4.1)

u is a null coordinate

To calculate r , the affine parameter, we note that t is an affine parameter for the metric within the square

brackets. Therefore $r = \int \Omega^2 dt + B(u, \theta, \phi)$ (4.2)

will be an affine parameter for (4.1)

B is constant along the null geodesic. Normally it would be taken so that $r = 0$ when $t = u$. However, in our case it will be more convenient to make it zero and define r as

$$r = \int_0^t \Omega^2 dt' \quad (4.3)$$

This means that surfaces of constant r are surfaces of constant t . This may seem rather odd, but it should be pointed out that the choice of B will not affect the asymptotic dependence of quantities. That is, if

$$f = O(r^{-n})$$

Then

$$f = O(r'^{-n}), \quad r' = r + B$$

It proves easier to perform the calculations with this choice of r but all results could be transformed back to a more normal coordinate system.

From (4.3) $r = A^2 \left[\frac{1}{4} \sinh^2 t - 2 \sinh t + \frac{3}{2} t \right]$ (4.4)

The matter in the universe is assumed to be dust so its energy tensor may be written

$$T_{ab} = \mu V_a V_b \quad (4.5)$$

For the undisturbed case, from Chapter 2

$$\mu = \frac{6A}{\Omega^3}$$

$$V_a = \Omega t_{;a}$$

$$V_a V^a = 1 \quad (4.6)$$

Now

$$\Omega = \sqrt{2} S + A - \frac{3A^2}{2\sqrt{2}} \frac{\log S}{S} + O(S^{-1}) \quad (4.7)$$

where $S^2 = r$

Therefore if we try to expand μ as a series in power of S the result will be very messy and will involve terms of the form $\frac{\log^n S}{S^n}$ *

*It should be pointed out that the expansions used will only be assumed to be valid asymptotically. They will not be assumed to converge at finite distances nor will the quantities concerned be assumed analytic. (see A. Erdelyi: Asymptotic Expansions - Dover)

This does not invalidate it as an asymptotic expansion but it makes it tedious to handle. For convenience therefore, we will perform the expansions in terms of $\Omega(r)$ which will be defined in general as the same function of r as it is in the undisturbed case. That is

$$\Omega = A(\cosh t - 1)$$

where
$$\tau = A^2 \left[\frac{1}{4} \sinh 2t - 2 \sinh t + \frac{3}{2} t \right] \quad (4.8)$$

then

$$\frac{d\Omega}{d\tau} = \frac{\sqrt{1 + \frac{2A}{\Omega}}}{\Omega}$$

$$= \frac{1}{\Omega} \left[1 + \frac{A}{\Omega} - \frac{A^2}{2\Omega^2} + \frac{A^3}{2\Omega^3} - \frac{5A^4}{8\Omega^4} + \frac{7A^5}{8\Omega^5} \dots \right] \quad (4.9)$$

For the third and fourth coordinates it is more convenient to use stereographic coordinates than spherical polars.

Since the matter is dust its energy-momentum tensor and hence the Ricci-tensor have only four independent components. We will take these as $\Lambda, \phi_{00}, \phi_{01}$.

(Since ϕ_{01} is complex it represents two components)

In terms of these the other components of the Ricci-tensor may be expressed as:

$$\phi_{11} = 3\Lambda + \frac{\phi_{01} \bar{\phi}_{01}}{\phi_{00}}$$

$$\phi_{22} = \frac{36\Lambda^2}{\phi_{00}} \left(1 + \frac{\phi_{01} \bar{\phi}_{01}}{6\Lambda \phi_{00}} \right)^2$$

$$\phi_{12} = \bar{\phi}_{21} = \frac{6\Lambda\phi_{01}}{\phi_{00}} \left(1 + \frac{\phi_{01}\bar{\phi}_{01}}{6\Lambda\phi_{00}} \right)^2$$

$$\phi_{02} = \bar{\phi}_{20} = \frac{\phi_{01}^2}{\phi_{00}} \quad (4.10)$$

For the undisturbed universe with the coordinate system given:

$$\Lambda = \frac{\mu}{24} = \frac{A}{4\Omega^3}$$

$$\phi_{00} = \frac{3A}{\Omega^5}$$

$$\phi_{11} = \frac{3A}{4\Omega^3}$$

$$\phi_{22} = \frac{3A}{4\Omega}$$

$$\phi_{01} = \phi_{02} = 0$$

(4.11)

Using these values and the fact that in the undisturbed universe all the $\psi^{\mu s}$ are zero, we may integrate equations (3. 10-50) to find the values of the spin coefficients for the unperturbed universe:

$$\rho = -\frac{2}{\Omega^2} - \frac{A}{\Omega^3} + \left(\frac{A^2}{2} - \frac{A^2}{2} e^{2u} \right) \Omega^{-4} - A^3 \left(\frac{1}{2} - e^{2u} \right) \Omega^{-5} + A^4 \left(\frac{5}{8} - \frac{7}{4} e^{2u} - \frac{1}{8} e^{4u} \right) \Omega^{-6} \dots$$

$$\sigma = \tau = \omega = \nu = \lambda = \chi^u = 0$$

$$\xi^3 = \frac{Se^u}{\Omega^2} - \frac{AS^e u}{\Omega^3} + A^4 (5SA^u + Se^{3u}) \Omega^{-4} + \dots$$

$$\alpha = -\bar{\beta} = \frac{1}{2} \frac{\bar{\nabla} S e^u}{\Omega^2} - \frac{1}{2} \frac{A \bar{\nabla} S e^u}{\Omega^3} + \dots$$

$$\mu = \frac{A}{2\Omega} - \frac{A^2}{4} (1 + e^{2u}) \Omega^{-2} + \frac{A^3}{4} (1 + 2e^{2u}) \Omega^{-3} + \dots$$

$$\gamma = -\frac{1}{2} - \frac{A}{2\Omega} + \frac{A^2}{4\Omega^2} - \frac{A^3}{4\Omega^3} + \dots$$

$$v = \frac{1}{2} \Omega^2 \quad (4.12)$$

5. Boundary Conditions

We wish to consider radiation in a universe that asymptotically approaches the undisturbed universe given above. ϕ_{00} and Λ will then have the values given above plus terms of smaller order. To determine this order and the order of ϕ_{01} and ψ_0 , there are two ways in which we may proceed. We may take the smallest orders that will permit radiation, that is $\psi_4 = O(r^{-1})$. Larger order terms than these in ϕ_{00} , Λ and ϕ_{01} turn out to have their u derivatives dependent only on themselves and not on the r^{-1} coefficient of ψ_4 , the radiation field. They are thus disturbances not produced by the radiation field and will not be considered.

Alternatively we may proceed by a method of successive

approximations. We take the undisturbed values of the spin coefficients and use them to solve the Bianchi Identities as field equations for the conformal tensor using the flat space boundary condition that $\psi_0 = O(r^{-5})$. Then substituting these ψ^s in equations (3.10 - 27) calculate the disturbances induced in the spin coefficients and substituting these back in the Bianchi Identities, calculate the disturbances in the ψ^s . Further iteration does not affect the orders of the disturbances.

Both these methods indicate that the boundary conditions should be:

$$\Lambda = \frac{A}{4\Omega^3} + O(\Omega^{-7}) \quad (5.1)$$

$$\phi_{00} = \frac{3A}{\Omega^5} + O(\Omega^{-9}) \quad (5.2)$$

$$\phi_{01} = O(\Omega^{-7}) \text{ (see next section)} \quad (5.3)$$

$$\psi_0 = O(\Omega^{-7}) \quad (5.4)$$

We also assume "uniform smoothness", that is:

$$\frac{\partial}{\partial x^c} \frac{\partial}{\partial x^d} \dots \frac{\partial}{\partial x^c} \Lambda = O(\Omega^{-7})$$

$$\frac{\partial \Lambda}{\partial \Omega} = \frac{-3A}{4\Omega^4} + O(\Omega^{-8})$$

etc.../

It will be shown that if these boundary conditions hold on one hypersurface ($u = \text{const.}$) they will hold on succeeding hypersurfaces and that these conditions are the most severe to permit radiation.

6 Integration

As Newman and Penrose, we begin by integrating the equations (3. 10 & 11)

$$\begin{aligned} D\rho &= \rho^2 + \sigma\bar{\sigma} + \phi_{00} \\ D\sigma &= 2\rho\sigma + \psi_0 \end{aligned}$$

where

$$\phi_{00} = \frac{3A}{4\sqrt{2}r^{\frac{5}{2}}} + O(r^{-3})$$

$$\psi_0 = \frac{\psi_0^0}{8\sqrt{2}r^{\frac{7}{2}}} + O(r^{-4})$$

Let

$$P = \begin{bmatrix} \rho & \sigma \\ \bar{\sigma} & \rho \end{bmatrix} \quad Q = \begin{bmatrix} \phi_{00} & \psi_0 \\ \bar{\psi}_0 & \phi_{00} \end{bmatrix}$$

$$\text{then } DP = P^2 + \varphi \quad (6.1)$$

$$\text{let } P = -(DY)Y^{-1} \quad (6.2)$$

$$\text{then } D^2 Y = -\varphi Y \quad (6.3)$$

$$\text{since } \int r \varphi dr < \infty$$

$$DY = F + O(1)$$

$$\text{where } F \text{ is constant} \quad (6.4)$$

$$Y = rF + O(r)$$

$$\text{However } \varphi = O(r^{-\frac{3}{2}})$$

$$\text{therefore } D^2 Y = -r\varphi F + O(r^{-\frac{3}{2}}) = O(r^{-\frac{3}{2}}) \quad (6.5)$$

$$\text{therefore } DY = F + O(r^{-\frac{1}{2}})$$

$$Y = rF + O(r^{\frac{1}{2}}) + E, \quad E \text{ is constant}$$

$$P = -r^{-1}I + O(r^{-\frac{3}{2}}) \quad (6.6)$$

if F is non-singular (The case F singular corresponds to asymptotically plane or cylindrical surfaces and will not be considered here).

$$\text{Thus } \rho = -r^{-1} + O(r^{-\frac{3}{2}}) = -2\Omega^{-2} + O(\Omega^{-3})$$

$$\sigma = O(r^{-\frac{3}{2}}) = O(\Omega^{-3}) \quad (6.7)$$

$$\text{Let } \rho = -2\Omega^{-2} + g\Omega^{-3}$$

$$\sigma = h\Omega^{-3} \quad (6.8)$$

where

$$g, h, = O(1)$$

Then using:

$$D = \frac{\partial}{\partial r} = \Omega^{-1} \left(1 + A\Omega^{-1} - \frac{A^2\Omega^2}{2} + \frac{A^3\Omega^3}{2} \dots \right) \frac{\partial}{\partial \Omega}$$

$$\frac{\partial g}{\partial \Omega} (\Omega + o(1)) = -A - g + o(\Omega^{-1})$$

Integrating,

$$g = \frac{1}{\Omega + o(1)} \left[a + \int \frac{(-A + o(\Omega^{-1})) (\Omega + o(1))}{\Omega + o(1)} d\Omega \right]$$

therefore

$$g = -A + o\left(\frac{\log \Omega}{\Omega}\right) \quad (6.9)$$

For

h ,

$$\frac{\partial h}{\partial \Omega} (\Omega + o(1)) = -h + o(\Omega^{-1})$$

therefore

$$h = o(\Omega^{-1}) \quad (6.10)$$

Repeat the process with

$$\begin{aligned} \rho &= -2\Omega^{-2} - A\Omega^{-3} + g\Omega^{-4} \\ \sigma &= h\Omega^{-4} \end{aligned} \quad (6.12)$$

where $g = O(\log \Omega)$

$$h = O(1)$$

then

$$\begin{aligned} \frac{\partial h}{\partial \Omega} (\Omega + O(1)) &= O(\Omega^{-1}) \\ h &= \sigma^0(u, x^e) + O(\Omega^{-1}) \end{aligned} \quad (6.13)$$

$$\frac{\partial g}{\partial \Omega} (-\Omega + O(1)) = O(\Omega^{-1} \log \Omega)$$

$$g = \rho^0(u, x^e) + O(\Omega^{-1} \log \Omega) \quad (6.14)$$

Unlike Newman and Unti, we cannot make ρ^0 zero by the transformation $\tau' = \tau - \rho^0$, since this would alter the boundary condition $\Lambda = \frac{A}{4\Omega^3} + \frac{\Delta^0}{\Omega^7}$

Continuing the above process we derive:

$$\begin{aligned} \rho &= -2\Omega^{-2} - A\Omega^{-3} + \rho^0\Omega^{-4} + \left(\frac{1}{2}A^2 - 2A\rho^0\right)\Omega^{-5} \\ &+ \left(-\frac{5}{4}A^4 + 4A^2\rho^0 - \frac{\rho^{0^2}}{2} - \frac{\sigma^0\bar{\sigma}^0}{2}\right)\Omega^{-6} + O(\Omega^{-7}) \end{aligned} \quad (6.15)$$

$$\sigma^0 = \sigma^0\Omega^{-4} - (2A\sigma^0 + \psi_0^0)\Omega^{-5} + \dots \quad (6.16)$$

To determine the asymptotic behaviour of $\psi, \alpha, \beta, \xi^4$ and ω we use the lemma proved by Newman and Penrose:

The $n \times n$ matrix B and the column vector b are given functions of x such that:

$$B = O(x^{-2}), \quad b = O(x^{-2}) \quad (6.17)$$

The $n \times n$ matrix A is independent of x and has no eigenvalue with positive real part. Any eigenvalue with vanishing real part is regular. Then all solutions of:

$$\frac{\partial}{\partial x} y = (Ax^{-1} + B)y + b \quad (6.18)$$

are bounded as $x \rightarrow \infty$. y is a column vector.

For reasons to be explained below, we will assume for the moment that

$$\begin{aligned} \phi_{0,1} &= O(\Omega^{-5}) \\ \frac{\partial}{\partial \Omega} \phi_{0,1} &= O(\Omega^{-6}) \\ \frac{\partial}{\partial x^0} \frac{\partial}{\partial x^i} \dots \phi_{0,1} &= O(\Omega^{-5}) \end{aligned} \quad (6.19)$$

We take as y the column vector

$$[\Omega^5 \psi, \Omega^2 A, \Omega^2 \bar{A}, \Omega^2 \beta, \Omega^2 \xi^3, \Omega^2 \bar{\xi}^3, \Omega^2 \xi^4, \Omega \bar{\xi}^4, \omega, \bar{\omega}] \quad (6.20)$$

By equations 3.51, 3.13, 3.14, 3.43, 3.44

$$A = \begin{vmatrix} -3 & 0 & 6A & 6A & 0 & 0 & 0 & 0 & 0 & 15A & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & & & & & & & & & & 0 \\ 0 & & & & & & & & & & 0 \\ 0 & & & & & & & & & & 0 \\ 0 & & & & & & & & & & 0 \\ 0 & & & & & & & & & & 0 \\ 0 & & & & & & & & & & 0 \\ 0 & & & & & & & & & & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & -1 & 0 & 0 & 0 & 0 & 0 & -2 & 0 \\ 0 & -1 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & -2 \end{vmatrix}$$

(6.21)

B and b are $O(\Omega^{-1})$ expressions involving

$$\rho, \sigma, \psi_0, \frac{\partial \psi_0}{\partial \Omega}, \frac{\partial \psi_0}{\partial x}, \phi_{01} \quad \text{and} \quad \frac{\partial \phi_{01}}{\partial \Omega}$$

Thus $\psi_1 = O(\Omega^{-5})$
 $\alpha, \beta, \xi^3, \xi^4 = O(\Omega^{-2})$
 $\omega = O(1)$

(6.22)

Since $\tau = \bar{\alpha} + \beta$
 $\tau = O(\Omega^{-2})$

Using this we integrate equation (3.12) by the same method as above. We obtain

$$\tau = \tau^0(u, x^i) \Omega^{-2} + O(\Omega^{-3}) \quad (6.23)$$

We may make a null rotation of the tetrad on each null

geodesic

$$\begin{aligned} L'_\mu &= L_\mu \\ N'_\mu &= N_\mu + a \bar{m}_\mu + \bar{a} m_\mu + a \bar{a} l_\mu \\ M'_\mu &= M_\mu + a L_\mu \end{aligned} \quad (6.24)$$

a is constant along the geodesic since the tetrad is parallelly transported.

By taking $a = -\frac{1}{2} \tau^0$ we may make $\tau'^0 = 0$

Under a null rotation

$$\phi'_{01} = \phi_{01} + a \phi_{00}$$

Thus until we have specified the null rotation we cannot

impose a boundary condition on ϕ_{01} more severe than

$\phi_{01} = O(\Omega^{-5})$. We will specify the null rotation by

$\tau^0 = 0$ and in that tetrad system will impose the boundary condition that $\phi_{01} = O(\Omega^{-7})$ and is uniformly smooth.

Then by using this condition on ϕ_{01} and $\tau = O(\Omega^{-3})$ by equation (3.44)

$$\omega = O(\Omega^{-1})$$

using this in equation (3.51)
 $\psi_1 = O(\Omega^{-6})$

then by equation (3.12) $\tau = O(\Omega^{-4})$

putting this back in equation (3.44)
 $\omega = O(\Omega^{-2} \log \Omega)$

by equation (3.51) $\psi_1 = O(\Omega^{-7} \log \Omega)$

by equation (3.12) $\tau = O(\Omega^{-5} \log \Omega)$

by equation (3.44) $\omega = \omega^0 \Omega^{-2} + O(\Omega^{-3} \log \Omega)$

by equation (3.51) $\psi_1 = O(\Omega^{-7})$ (6.25)

by equation (3.12) $\tau = O(\Omega^{-5})$ (6.26)

by equation (3.44) $\omega = \omega^0 \Omega^{-2} + O(\Omega^{-3})$ (6.27)

By differentiating the equations used with respect to x^i ,
 one may show that $\psi_1, \alpha, \beta, \tau, \xi^L, \omega$ are
 uniformly smooth.

Adding equations 3.53 and: 3.60 :-

$$\begin{aligned} \delta \psi_1 - D \psi_2 + D \phi_{11} - \delta \phi_{01} + D1 &= \lambda \psi_0 - 3\rho \psi_2 + 2\alpha \psi_1 - \mu \phi_{00} - 2\bar{\alpha} \phi_{10} \\ &+ 2\rho \phi_{11} + \sigma \phi_{20} \end{aligned} \quad (6.28)$$

We may use the lemma again with y

$$y = \begin{bmatrix} \Omega^4 \psi_2 \\ \Omega^2 \lambda \\ \Omega \mu \end{bmatrix}$$

By equations 3.16; 3.17; 6.28.

$$A = \begin{bmatrix} -2 & 0 & 3A \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

B and b are

$$O(\Omega^{-2})$$

Therefore

$$\psi_2 = O(\Omega^{-4})$$

$$\lambda = O(\Omega^{-2})$$

(6.29)

$$\mu = O(\Omega^{-1})$$

and are uniformly smooth

From (6.29) and (3.17) we may show

$$\mu = \frac{A}{2} \Omega^{-1} + O(\Omega^{-2} \log \Omega)$$

using this in (6.28)

$$\psi_2 = O(\Omega^{-5} \log \Omega)$$

then by (3.17)

$$\mu = \frac{A}{2} \Omega^{-1} + \mu^0(u, x') \Omega^{-2} + O(\Omega^{-3} \log \Omega) \quad (6.30)$$

and $\psi_2 = O(\Omega^{-5})$

Integrating the radial equations 3.13, 3.14, 3.15, 3.16, 3.43, 3.45
3.46.

$$\alpha = \alpha^0 \Omega^{-2} - A \alpha^0 \Omega^{-3} + \frac{1}{2} (3A^2 \alpha^0 - \rho^0 \alpha^0 + \bar{\alpha}^0 \sigma^0) \Omega^{-4} + O(\Omega^{-5}) \quad (6.31)$$

$$\beta = \beta^0 \Omega^{-2} - A \beta^0 \Omega^{-3} + \frac{1}{2} (3A^2 \beta^0 - \rho^0 \beta^0 + \bar{\beta}^0 \sigma^0) \Omega^{-4} + O(\Omega^{-5}) \quad (6.32)$$

$$\gamma = \gamma^0 - \frac{A}{2} \Omega^{-1} + \frac{A^2}{4} \Omega^{-2} + O(\Omega^{-3}) \quad (6.33)$$

$$\lambda = \lambda^0 \Omega^{-2} - A (\lambda^0 + \frac{\sigma^0}{2}) \Omega^{-3} + O(\Omega^{-4}) \quad (6.34)$$

$$\xi^{\mu} = \xi^{\mu 0} \Omega^{-2} - A \Omega^3 \xi^{\mu 0} + \frac{1}{2} (3A^2 \xi^{\mu 0} - \rho^0 \xi^{\mu 0} - \bar{\xi}^{\mu 0} \sigma^0) \Omega^{-4} + O(\Omega^{-5}) \quad (6.35)$$

$$X^{\mu} = X^{\mu 0} + O(\Omega^{-5}) \quad (6.36)$$

$$U = -\frac{1}{2} (\gamma^0 + \bar{\gamma}^0) \Omega^2 + A (1 + \gamma^0 + \bar{\gamma}^0) \Omega - \frac{3}{2} A^2 \\ \times (1 + \gamma^0 + \bar{\gamma}^0) \omega_J \Omega + U^0 + O(\Omega^{-1}) \quad (6.37)$$

Adding equation (3.55) + 2 x (3.59)

$$\delta \psi_2 - \mathcal{D} \psi_3 + \mathcal{D} \phi_{21} - \delta \phi_{20} + 2 \delta \Pi = 2 \lambda \psi_1 - 2 \rho \psi_3 \\ - \frac{1}{3} (\bar{\mu} + 2 \mu) \phi_{10} + 2 (\beta - \bar{\alpha}) \phi_{20} + 2 \rho \phi_{21} \quad (6.38)$$

Therefore $\psi_3 = \psi_3^0 \Omega^{-4} + O(\Omega^{-5}) \quad (6.39)$

By equation (3.18)

$$V = V^0 - \frac{1}{2} \psi_3^0 \Omega^{-2} + O(\Omega^{-3}) \quad (6.40)$$

By the orthonormality relations (2.2).

$$\begin{aligned} g^{2i} &= X^i - (\xi^i \bar{\omega} + \bar{\xi}^i \omega) \\ &= X^{i0} + O(\Omega^{-4}) \\ g^{ij} &= -(\xi^i \bar{\xi}^j + \bar{\xi}^i \xi^j) \quad i, j = 3, 4 \\ &= -(\xi^{i0} \bar{\xi}^{j0} + \bar{\xi}^{i0} \xi^{j0}) (\Omega^{-4} 2A \Omega^{-5}) + O(\Omega^{-6}) \quad (6.41) \end{aligned}$$

By making the coordinate transformation

$$\begin{aligned} u' &= u \\ r' &= r \\ x'^i &= x^i + C^i(u, x^i) \end{aligned}$$

where

$$\begin{aligned} C_{,1}^3 &= -X^{30} (1 + C_{,3}^3) - X^{40} C_{,4}^3 \\ C_{,1}^4 &= -X^{40} (1 + C_{,4}^4) - X^{30} C_{,3}^4 \quad (6.42) \\ X'^{i0} &= 0 \end{aligned}$$

We still have the coordinate freedom

$$x^i = D^i(x^j)$$

We may use this to reduce the leading term of $g^{ij} (i, j = 3, 4)$ to a conformally flat metric (c.f. Newman and Unti), that is:

$$g^{ij} = -2P\bar{P}\delta^{ij} (\Omega^{-4} - 2A\Omega^{-5}) + O(\Omega^{-6}) \quad (6.43)$$

where

$$\xi^{30} = -i \xi^{40} = P(u, x^i).$$

7. Non-radial Equations

By comparing coefficients of the various powers of Ω in the non-radial equations of §3, relations between the integration constants of the radial equations may be obtained:

In equation 3.23 the term in Ω^{-1} is

$$-\frac{3}{4}A(\gamma^0 + \bar{\gamma}^0) - \frac{3}{4}A$$

therefore

$$\gamma^0 + \bar{\gamma}^0 = -1$$

(7.1)

therefore by (6.37)

$$U = \frac{1}{2}\Omega^2 + U^0 + O(\Omega^{-1})$$

(7.2)

In equation (3.50), the constant term is

$$V^0$$

therefore

$$V^0 = 0$$

(7.3)

By the Ω^{-2} term in (3.47)

$$P - P_{,i} = (\bar{\gamma}^0 - \gamma^0)P$$

By the Ω^{-3} term

$$P - P_{,i} = 0$$

therefore

$$\gamma^0 = \bar{\gamma}^0 = -\frac{1}{2}$$

(7.4).

$$P = S(x^{\mu}) e^u$$

if $S = |S| e^{i\phi}$

By making a spatial rotation of the tetrad

we make S real. We take $S = \frac{1}{\sqrt{2}} \left(1 + \frac{1}{4} (x^2 + x^4) \right)$, the stereographic project factor for a 2-sphere.

By the Ω^{-4} term in equation (3.48)

$$\begin{aligned} \alpha^0 &= \frac{1}{2} \bar{\nabla} S e^u \\ \beta^0 &= -\frac{1}{2} \nabla S e^u \end{aligned} \quad (7.5)$$

where $\nabla = \frac{\partial}{\partial x^3} + i \frac{\partial}{\partial x^4}$

By the Ω^{-4} term in (3.21)

$$\begin{aligned} \frac{1}{2} e^{2u} (S \nabla \bar{\nabla} S + S \bar{\nabla} \nabla S) &= -2\mu^0 - \frac{A^2}{2} + (\nabla P)(\bar{\nabla} P) \\ \mu^0 &= -\frac{A^2}{4} - \frac{S^2}{2} \nabla \bar{\nabla} \log S e^{2u} \\ &= -\frac{A^2}{4} [1 + e^{2u}] \end{aligned} \quad (7.6)$$

By (3.59) (3.60) (3.61)

$$\frac{\partial \phi_{01}}{\partial u} = O(\Omega^{-7})$$

$$\frac{\partial}{\partial u} \phi_{00} = O(\Omega^{-7})$$

$$\frac{\partial}{\partial u} \Lambda = H\Omega^{-5} + O(\Omega^{-6} \log \Omega) \quad (7.7)$$

where

$$H = -\frac{A^2}{8} + \frac{3U^0}{4} + \frac{\rho^0}{4} + \frac{A^2 e^{2u}}{8}$$

By making a coordinate transformation*

$$u' = u - \int_{u_0}^u H du'' \quad (7.8)$$

$$H' = 0$$

therefore

$$U^0 = \frac{A^2}{6} - \frac{A^2 e^{2u}}{6} - \frac{\rho^0}{3}$$

By the Ω^{-4} term in (3.26)

$$\rho_{,11}^0 - \rho^0 + 4U^0 = -\frac{A^2}{2}(1 + e^{2u}) \quad (7.9)$$

therefore

$$\rho_{,11}^0 - \frac{7}{3}\rho^0 + \frac{7}{6}A^2 - \frac{1}{6}A^2 e^{2u} = 0$$

$$\rho^0 = \frac{A^2}{2}(1 - e^{2u}) + C(x^i) e^{\frac{7}{3}u} \quad (7.10)$$

*This transformation does not upset the boundary conditions on the hypersurface $u = u^0$

By the Ω^{-4} term in (3.25)

$$\lambda^0 = \frac{1}{2} (\bar{\sigma}_{,1}^0 - \bar{\sigma}^0) \quad (7.11)$$

By the Ω^{-4} term in (3.22)

$$\psi_3^0 = (\bar{\sigma}_{,1}^0 - \bar{\sigma}^0) e^u \nabla S - \frac{1}{2} e^u S \nabla (\bar{\sigma}_{,1}^0 - \bar{\sigma}^0) \quad (7.12)$$

By the Ω^{-2} term in (3.50)

$$2\omega^0 - \omega_{,1}^0 = \frac{1}{2} \bar{\psi}_3^0$$

$$\omega^0 = \frac{e^u}{4} (S \bar{\nabla} \bar{\sigma}^0 - 2 \bar{\sigma}^0 \bar{\nabla} S) + K(x^i) e^{2u} \quad (7.13)$$

By the Ω^{-6} term in (3.20)

$$e^u S \nabla \rho^0 + 4\omega^0 - e^u S \bar{\nabla} \bar{\sigma}^0 = -2\bar{\sigma}^0 \bar{\nabla} S e^u$$

therefore $C = K = 0$

(7.14)

therefore $U^0 = 0$

(7.15)

$$\rho^0 = \frac{A^2}{2} (1 - e^{2u})$$

(7.16)

Using (7.6)(7.16), in (6.28)

$$\psi_2 = O(\Omega^{-6} \log \Omega)$$

Then by (3.17)

$$\begin{aligned} \mu &= \frac{A}{2} \Omega^{-1} - \frac{A^2}{4} (1 + e^{2u}) \Omega^{-2} + \frac{A^3}{2} \left(\frac{1}{2} + e^{2u} \right) \Omega^{-3} \\ &\quad + O(\Omega^{-4} \log \Omega) \end{aligned} \quad (7.17)$$

Using this in (6.28)

$$\psi_2 = O(\Omega^{-6}) \quad (7.18)$$

By (3.51), (3.59), (3.60), (3.61)

$$\frac{\partial}{\partial u} \phi_{01} = O(\Omega^{-7})$$

$$\frac{\partial}{\partial u} \phi_{00} = O(\Omega^{-9})$$

$$\frac{\partial}{\partial u} \Lambda = O(\Omega^{-7})$$

$$\frac{\partial}{\partial u} \psi_0 = O(\Omega^{-7})$$

(7.19)

Therefore if the boundary conditions (5.1-4) hold on one null hypersurface, they will hold on succeeding hypersurfaces.

By (3.57)

$$\psi_4 = O(\Omega^{-2})$$

(7.20)

The "peeling off" behaviour is therefore:

$$\psi_4 = O(r^{-1})$$

$$\psi_3 = O(r^{-2})$$

$$\psi_2 = O(r^{-3})$$

$$\psi_1 = O(r^{-\frac{7}{2}})$$

$$\psi_0 = O(r^{-\frac{7}{2}})$$

(7.21)

As mentioned before, this asymptotic behaviour is independent of the zero of γ and will hold for any affine parameter r .

To perform the remaining integrations we will assume for definiteness:

$$\begin{aligned}\phi_{01} &= \phi_{01}^0 \Omega^{-7} + \phi_{01}^1 \Omega^{-8} + o(\Omega^{-9}) \\ \psi_0 &= \psi_0^0 \Omega^{-7} + \psi_0^1 \Omega^{-8} + o(\Omega^{-9}) \\ \Lambda &= \Lambda \Omega^{-3} + \Lambda^0 \Omega^{-7} + o(\Omega^{-8}) \\ \phi_{00} &= 3A \Omega^{-5} + \phi_{00}^0 \Omega^{-9} + o(\Omega^{-10})\end{aligned} \quad (7.22)$$

Then:

$$\begin{aligned}\rho &= -2\Omega^{-2} - A\Omega^{-3} + \frac{A^2}{2}(1 - e^{2u})\Omega^{-4} - \frac{A^3}{2}(1 - 2e^{2u})\Omega^{-5} \\ &\quad + \left[A^4 \left(\frac{5}{8} - \frac{7}{8}e^{2u} - \frac{1}{8}e^{4u} \right) - \frac{\sigma^0 \bar{\sigma}^0}{2} \right] \Omega^{-6} \\ &\quad + \frac{1}{3} \left[A^5 \left(-\frac{37}{8} + 8e^{2u} + 8e^{4u} + 2e^{6u} \right) - \phi_{00}^0 \right. \\ &\quad \left. + \bar{\sigma}^0 (3A\sigma^0 + \psi_0^0) + \sigma^0 (3A\bar{\sigma}^0 + \bar{\psi}_0^0) \right] \Omega^{-5} \\ &\quad + o(\Omega^{-8})\end{aligned} \quad (7.23)$$

$$\sigma = \sigma^0 \Omega^{-4} - (2A\sigma^0 + \psi_0^0) \Omega^{-5} + o(\Omega^{-6}) \quad (7.24)$$

$$\tau = -\frac{1}{3}(\phi_{01}^0 + \psi_1^0) \Omega^{-5} + o(\Omega^{-6} \log \Omega) \quad (7.26)$$

$$\omega = \frac{e^u}{4} (S \bar{\nabla} \sigma^0 - 2 \sigma^0 \bar{\nabla} S) \Omega^{-2} - \left[\frac{A e^u}{4} (S \bar{\nabla} \sigma^0 - 2 \sigma^0 \bar{\nabla} S) \right. \\ \left. + \frac{1}{3} (\phi_{0i}^0 + \psi_i^0) \right] \Omega^{-3} + O(\Omega^{-4} \log \Omega) \quad (7.26)$$

$$\gamma = -\frac{1}{2} - \frac{A}{2} \Omega^{-1} + \frac{A^2}{4} \Omega^{-2} - \frac{A^3}{4} \Omega^{-3} + \frac{1}{4} \left(\frac{5}{4} A^4 - \psi_2^0 \right) \Omega^{-4} \\ + O(\Omega^{-5}) \quad (7.27)$$

$$\mu = \frac{A}{2} \Omega^{-1} - \frac{A^2}{4} (1 + e^{2u}) \Omega^{-2} + \frac{A^3}{4} (1 + 2e^{2u}) \Omega^{-3} \\ - \frac{1}{2} \left[A^4 \left(\frac{5}{8} + \frac{7}{4} e^{2u} + \frac{1}{8} e^{4u} \right) + \frac{6}{2} (\bar{\sigma}_{,1}^0 - \bar{\sigma}^0) + \psi_2^0 \right] \\ \times \Omega^{-4} + O(\Omega^{-5}) \quad (7.28)$$

$$\lambda = \frac{1}{2} (\bar{\sigma}_{,1}^0 - \bar{\sigma}^0) \Omega^{-2} - \frac{A \bar{\sigma}_{,1}^0}{2} \Omega^{-3} + O(\Omega^{-4}) \quad (7.29)$$

$$V = -\frac{1}{2} ((\bar{\sigma}_{,1}^0 - \bar{\sigma}^0) e^u \nabla S - \frac{1}{2} e^u S \nabla (\bar{\sigma}_{,1}^0 - \bar{\sigma}^0)) \Omega^{-2} \\ + O(\Omega^{-3}) \quad (7.30)$$

$$\alpha = \frac{1}{2} e^u \bar{\nabla} S \Omega^{-2} - \frac{1}{2} A e^u \bar{\nabla} S \Omega^{-3} \\ + \frac{1}{4} \left[A^2 e^u \nabla S \left(\frac{5}{2} + \frac{1}{2} e^u \right) + e^u (\nabla S) \bar{\sigma}^0 \right] \Omega^{-4} \\ + O(\Omega^{-5}) \quad (7.31)$$

$$\xi^3 = e^u S \Omega^{-2} - A e^u S \Omega^{-3} + \frac{1}{2} [A^2 e^u S (\frac{S}{2} + \frac{1}{2} e^u) - e^u S \sigma^0] \Omega^{-4} + o(\Omega^{-5}) \quad (7.32)$$

$$U = \frac{1}{2} \Omega^2 - \frac{1}{8} (\psi_2^0 + \bar{\psi}_2^0) \Omega^{-2} + o(\Omega^{-3}) \quad (7.33)$$

$$X^3 = \frac{e^u S}{15} [\phi_{0i}^0 + \bar{\phi}_{0i}^0 + \psi_1^0 + \bar{\psi}_1^0] \Omega^{-5} + o(\Omega^{-6} \log \Omega) \quad (7.34)$$

$$\begin{aligned} \psi_u^0 = & \left[-\bar{\sigma}^0 + \frac{3}{2} \sigma_{,1}^0 + -\frac{1}{2} \sigma_{,11}^0 \right] \Omega^{-2} + \left[-\frac{\bar{\sigma}^0}{2} - \frac{3}{4} \bar{\sigma}_{,11}^0 \right. \\ & \left. + \frac{1}{2} \bar{\sigma}_{,11}^0 \right] \Omega^{-3} + \left[A^2 (\bar{\sigma}^0 + \frac{3}{4} \bar{\sigma}_{,11}^0 - \frac{3}{8} \bar{\sigma}_{,11}^0) \right. \\ & + \frac{A^2}{4} (-\bar{\sigma}^0 + \frac{3}{2} \bar{\sigma}_{,11}^0 - \frac{1}{2} \bar{\sigma}_{,11}^0) e^{2u} - e^{2u} \bar{\nabla} (S \\ & \times (\bar{\sigma}_{,11}^0 - \bar{\sigma}^0)^{\frac{3}{2}} \nabla (S (\bar{\sigma}_{,11}^0 - \bar{\sigma}^0)^{\frac{1}{2}})) \\ & \left. + \frac{3A}{8} \bar{\psi}_0^0 \right] \Omega^{-4} + o(\Omega^{-5}) \quad (7.35) \end{aligned}$$

$$\begin{aligned} \psi^3 = & \left[e^u (\bar{\sigma}_{,11}^0 - \bar{\sigma}^0) \nabla S - e^u S \nabla (\bar{\sigma}_{,11}^0 - \bar{\sigma}^0) \right] \Omega^{-4} \\ & + \left[A e^u \left(-2 (\bar{\sigma}_{,11}^0 - \frac{11}{8} \bar{\sigma}^0) \nabla S - S \nabla (\bar{\sigma}_{,11}^0 - \frac{11}{8} \bar{\sigma}^0) \right) \right. \\ & \left. + \frac{1}{2} \bar{\phi}_{0i}^0 \right] \Omega^{-5} + o(\Omega^{-6}) \quad (7.36) \end{aligned}$$

$$\begin{aligned} \psi_2 = & \psi_2^0 \Omega^{-6} + \left[-\frac{3}{2} A \psi_2^0 - e^u S^2 \bar{\nabla} \left(\frac{\psi_1^0 - \phi_{01}^0}{S} \right) \right. \\ & + \frac{1}{2} \psi_0^0 (\bar{\sigma}_{11}^0 - \bar{\sigma}^0) + \frac{3A}{4} \sigma^0 \bar{\sigma}_{11}^0 - \frac{3A}{2} \sigma^0 \bar{\sigma}^0 \\ & \left. + 16A^0 \right] \Omega^{-7} + O(\Omega^{-8} \log \Omega) \end{aligned} \quad (7.37)$$

where:

$$\begin{aligned} \psi_2^0 - \bar{\psi}_2^0 = & \left[e^{2u} \left(\frac{S^2}{2} \bar{\nabla} \bar{\nabla} \sigma^0 - S (\bar{\nabla} S) (\bar{\nabla} \sigma^0) \right. \right. \\ & \left. \left. + \sigma^0 (\bar{\nabla} S)^2 - \sigma^0 S \bar{\nabla} \bar{\nabla} S - \frac{1}{2} \sigma^0 \bar{\sigma}_{11}^0 \right) \right] - c.c. \end{aligned}$$

$$\psi_2^0 + \bar{\psi}_2^0 \text{ is undetermined} \quad (7.38)$$

$$\psi_1 = \psi_1^0 \Omega^{-7} + \psi_1^1 \Omega^{-8} \log \Omega + \psi_1^2 \Omega^{-8} + O(\Omega^{-9} \log \Omega)$$

where

$$\begin{aligned} \psi_1^0 = & e^u \left[S \bar{\nabla} \left(\psi_0^0 + \frac{15}{4} \sigma^0 \right) - \left(2 \psi_0^0 + \frac{15}{2} \sigma^0 \right) \bar{\nabla} S \right] \\ & - 3 \phi_{01}^0 \end{aligned} \quad (7.39)$$

$$\begin{aligned} \psi_1^1 = & -e^u (S \bar{\nabla} - 2 \bar{\nabla} S) (5A \psi_0^0 - \psi_0^1 + 15A \sigma^0) \\ & + 4 \phi_{01}^1 \end{aligned} \quad (7.40)$$

$$\psi_1^2 \text{ is undetermined}$$

$$\delta_4 \quad (3.54)$$

$$\psi_{1,1}^1 = 3 \psi_1^1 \quad (7.41)$$

Thus the ω derivative of ψ' , depends only on itself and not on the radiation field. It therefore represents a type of disturbance unconnected with radiation. If it is zero on one hypersurface, it will remain zero. In this case it is possible to continue the expansions of all quantities in negative powers of Ω without any log terms appearing.

8. The Asymptotic Group

The metric has the form:

$$\begin{aligned} g'' &= g'^3 = g'^4 = 0, & g'^2 &= 1 \\ g^{22} &= \Omega^2 + O(\Omega^{-2}) \\ g^{2i} &= O(\Omega^{-4}) \\ g^{ij} &= -2P^2 g^{ij} \Omega^{-4} + 2AP^2 g^{ij} \Omega^{-5} + O(\Omega^{-6}) \end{aligned}$$

The asymptotic group is the group of coordinate transformations that leave the form of the metric and of the boundary conditions unchanged. It can be derived most simply by considering the corresponding infinitesimal transformations:

$$x'^{\mu} = x^{\mu} + \epsilon K^{\mu}(x^{\nu}) \quad (8.1)$$